

BETTER THAN A CPU

Where do GPUs beat CPUs? Where does Verilog beat C++?
That question is the core theme of this ASIC hackathon.

Submissions should tackle problems in ways that are fundamentally better than a CPU implementation. “Better” might mean lower power, higher throughput, improved determinism, reliability, or exploiting massive parallelism... things general-purpose CPUs struggle with.

As an example, consider a parallel matrix multiplier: a classic hardware design that can exploit concurrency at a scale a CPU simply can't, delivering performance and efficiency that software alone can't reasonably approach.

- Designs must fit within Tiny Tapeout constraints:
- Maximum size: (INFINITE) (16 tiles is max for physical tapeout)
- Maximum I/O frequency: ~33-66 MHz

These limits are intentional. The challenge is to ask: even with these constraints, where does custom digital hardware still win? Think parallelization, streaming architectures, state machines, search and filtering, DSP, and classic FPGA-style accelerators.

Look for problems where hardware shines, and most importantly, have fun!

NOTE:

All submissions must be accompanied by an in-person presentation delivered to the judges explaining your design, teams will then have to answer an onslaught of questions asked by the judges, expect them to be hard.

More Details

The competition is split into two divisions: Junior and Senior.

Junior Division

The Junior Division is open only to first-year or non-ECE or non-EngSci students.

Junior submissions will be synthesized and implemented on an FPGA, specifically the DE1-SoC, to simplify the hardware bring-up and verification process. Competitors are encouraged to take advantage of the board's built-in resources, including on-chip memory, ADCs, and other available peripherals.

The focus in this division is learning, experimentation, and demonstrating how hardware acceleration can outperform CPU-based approaches.

Senior Division

The Senior Division is open to all participants.

Senior submissions must be fully synthesizable Tiny Tapeout designs that meet all size and interface constraints. Because fabricated ICs can take months to return, designs should emphasize safety, robustness, and correctness. A strong submission is one that is carefully verified and unlikely to require re-fabrication.

This division rewards thoughtful hardware architecture, reliability, and designs that fully embrace the strengths (and responsibilities) of custom silicon.

Example Projects

Tiny Binarized Neural Network (BNN) chip

https://www.tinytapeout.com/chips/ttsky25a/tt_um_BNN

A binarized neural network is a perfect “Verilog beats C++” example because it turns inference into tons of cheap parallel bit operations (XNOR + popcount style math) that can be done every cycle in hardware, while a CPU has to execute the same work as a long sequence of instructions with far worse energy-per-inference.

Monte Carlo π Integrator

https://tinytapeout.com/chips/tt03/regymm_mcpi

Monte Carlo is naturally parallel, and this project demonstrates that even in a small ASIC you can build a pipeline that generates random samples and counts hits nonstop, achieving a much higher “trials per second per watt” than a CPU doing the same thing sequentially with random-number generation and branching overhead.

GPS C/A PRN Generator

https://tinytapeout.com/chips/tt03/adamgreig_tt02_gps_ca_prn

This is a textbook “hardware shines” workload: generating GPS PRN sequences is essentially bit-accurate shift-register logic, so Verilog can produce one (or more) correct chips every clock with perfect timing, while a CPU implementation wastes cycles on bit manipulation and can struggle to match the same deterministic real-time behavior.